Project Structure (Professional MVC)

```
practicproject/
|
 ├── app/
|   ├── controllers/
|   |   └── StudentsController.php
|   |
|   ├── models/
|   |   └── Student.php
|   |
|   ├── views/
|   |   └── students/
|   |           ├── index.php
|   |           ├── create.php
|   |           ├── edit.php
|   |
|   └── core/
|           ├── Router.php
|           ├── Database.php
|
 ├── public/
|   └── index.php   (Entry point)
|
└── .htaccess
```

2. Clean URL (.htaccess)

```
RewriteEngine On

RewriteCond %{REQUEST_FILENAME} !-d
RewriteCond %{REQUEST_FILENAME} !-f

RewriteRule ^(.+)$ public/index.php?url=$1 [QSA,L]
```

Routes

localhost/practicproject/students
localhost/practicproject/students/create
localhost/practicproject/students/edit/1

3. Entry Point (public/index.php)

```
<?php


require_once "../app/core/Router.php";


$url = isset($_GET['url']) ? $_GET['url'] : 'students';


$url = explode('/', filter_var(trim($url, '/'), FILTER_SANITIZE_URL));


$controllerName = ucfirst($url[0]) . "Controller";

$method = isset($url[1]) ? $url[1] : "index";

$params = array_slice($url, 2);


require_once "../app/controllers/$controllerName.php";


$controller = new $controllerName();
```

```
call_user_func_array([$controller, $method], $params);
```

4. Database Connection (core/Database.php)

```
<?php

class Database {
    private $host = "localhost";
    private $db = "test1";
    private $user = "root";
    private $pass = "";

    public function connect() {
        return new PDO("mysql:host=$this->host;dbname=$this->db", $this->user, $this->pass);
    }
}
```

5. Model (models/Student.php)

```
<?php
require_once "../app/core/Database.php";

class Student {
```

```
private $db;

public function __construct() {
    $this->db = (new Database())->connect();
}

public function getAll() {
    return $this->db->query("SELECT * FROM students")->fetchAll(PDO::FETCH_ASSOC);
}

public function find($id) {
    $stmt = $this->db->prepare("SELECT * FROM students WHERE id=?");
    $stmt->execute([$id]);
    return $stmt->fetch(PDO::FETCH_ASSOC);
}

public function create($data) {
    $stmt = $this->db->prepare("INSERT INTO students (name, email) VALUES (?, ?)");
    return $stmt->execute([$data['name'], $data['email']]);
}

public function update($id, $data) {
    $stmt = $this->db->prepare("UPDATE students SET name=?, email=? WHERE id=?");
    return $stmt->execute([$data['name'], $data['email'], $id]);
}
```

```
    public function delete($id) {
        $stmt = $this->db->prepare("DELETE FROM students WHERE id=?");
        return $stmt->execute([$id]);
    }
}
```

6. Controller (controllers/StudentController.php)

```
<?php
require_once "../app/models/Student.php";

class StudentsController {

    private $student;

    public function __construct() {
        $this->student = new Student();
    }

    // READ
    public function index() {
        $students = $this->student->getAll();
        require "../app/views/students/index.php";
    }

    // CREATE FORM
    public function create() {
```

```
    require "../app/views/students/create.php";
}

// STORE
public function store() {
    $this->student->create($_POST);
    header("Location: /practicproject/students");
}

// EDIT FORM
public function edit($id) {
    $student = $this->student->find($id);
    require "../app/views/students/edit.php";
}

// UPDATE
public function update($id) {
    $this->student->update($id, $_POST);
    header("Location: /practicproject/students");
}

// DELETE
public function delete($id) {
    $this->student->delete($id);
    header("Location: /practicproject/students");
}
```

}

7. Views

index.php (Read All)

```
<!DOCTYPE html>

<html>

<head>

    <title>Students</title>


    <!-- Bootstrap CDN -->

    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

</head>


<body class="bg-light">


<div class="container mt-5">


    <div class="d-flex justify-content-between mb-3">

        <h2 class="text-primary">Students List</h2>

        <a href="/practicproject/students/create" class="btn btn-success">+ Add Student</a>

    </div>


    <div class="card shadow">

        <div class="card-body">


            <table class="table table-bordered table-hover text-center">
```

```
<thead class="table-dark">
  <tr>
    <th>ID</th>
    <th>Name</th>
    <th>Email</th>
    <th width="200">Actions</th>
  </tr>
</thead>

<tbody>
<?php foreach($students as $s): ?>
  <tr>
    <td><?= $s['id'] ?></td>
    <td><?= $s['name'] ?></td>
    <td><?= $s['email'] ?></td>
    <td>
      <a href="/practicproject/students/edit/<?= $s['id'] ?>" class="btn btn-warning
btn-sm">Edit</a>

      <a href="/practicproject/students/delete/<?= $s['id'] ?>"
        class="btn btn-danger btn-sm"
        onclick="return confirm('Are you sure?')">
        Delete
      </a>
    </td>
  </tr>
```

```
            <?php endforeach; ?>
            </tbody>

        </table>

      </div>
    </div>

</div>

</body>
</html>
```

View -> create.php

```
<!DOCTYPE html>
<html>
<head>
    <title>Add Student</title>

    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>

<body class="bg-light">

<div class="container mt-5">
```

```
<div class="card shadow">
    <div class="card-header bg-success text-white">
        <h4>Add New Student</h4>
    </div>

    <div class="card-body">

        <form method="POST" action="/practicproject/students/store">

            <div class="mb-3">
                <label class="form-label">Student Name</label>
                <input type="text" name="name" class="form-control" placeholder="Enter name"
required>
            </div>

            <div class="mb-3">
                <label class="form-label">Email Address</label>
                <input type="email" name="email" class="form-control" placeholder="Enter email"
required>
            </div>

            <button type="submit" class="btn btn-success">Save</button>
            <a href="/practicproject/students" class="btn btn-secondary">Back</a>

        </form>
```

```
        </div>
    </div>

</div>

</body>
</html>
```

View -> edit.php

```
<!DOCTYPE html>
<html>
<head>
    <title>Edit Student</title>

    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>

<body class="bg-light">

<div class="container mt-5">

    <div class="card shadow">
        <div class="card-header bg-warning">
            <h4>Edit Student</h4>
        </div>
```

```
    <div class="card-body">

      <form method="POST" action="/practicproject/students/update/<?= $student['id'] ?>">

        <div class="mb-3">
          <label class="form-label">Student Name</label>
          <input type="text" name="name" class="form-control" value="<?= $student['name']
?>" required>
        </div>

        <div class="mb-3">
          <label class="form-label">Email Address</label>
          <input type="email" name="email" class="form-control" value="<?=
$student['email'] ?>" required>
        </div>

        <button type="submit" class="btn btn-warning">Update</button>
        <a href="/practicproject/students" class="btn btn-secondary">Back</a>

      </form>

    </div>
  </div>

</div>

</body>
```

```
</html>
```

8. Database Table

```
CREATE TABLE students (
   id INT AUTO_INCREMENT PRIMARY KEY,
   name VARCHAR(100),
   email VARCHAR(100)
);
```

File Explaination code

✅ **Code .htaccess**

```
RewriteEngine On

RewriteCond %{REQUEST_FILENAME} !-d
RewriteCond %{REQUEST_FILENAME} !-f

RewriteRule ^(.+)$ public/index.php?url=$1 [QSA,L]
```

🔍 **Step-by-Step Explanation**

🔹 **1. RewriteEngine On**

👉 Iska matlab:

URL rewriting system ON karo

📌 Agar ye line nahi hogi:
❌ baaki rules kaam nahi karenge

🔹 **2. RewriteCond %{REQUEST_FILENAME} !-d**

👉 RewriteCond = condition lagana

👉 %{REQUEST_FILENAME} = jo URL user ne request kiya

👉 !-d = "NOT a directory"

📌 Matlab:

agar requested path **folder nahi hai** tab next rule apply karo

---

🔹 **3. RewriteCond %{REQUEST_FILENAME} !-f**

👉 !-f = "NOT a file"

📌 Matlab:

agar requested path **file nahi hai** tab next rule apply karo

---

🎯 **In dono conditions ka purpose**

👉 Ye ensure karta hai:

Case	Kya hoga
Real file exist (image, css)	Direct open
Real folder exist	Direct open
Kuch exist nahi	MVC routing chalegi

🔹 **4. RewriteRule ^(.+)$ public/index.php?url=$1 [QSA,L]**

👉 Ye main rule hai 🔥

🔸 **^(.+)$**

👉 Ye regex (pattern) hai

Symbol	Meaning
^	start
(.+)	jo bhi likha hai (full URL)
$	end

📌 Matlab:

poora URL capture karo

🔸 **$1**

👉 Ye captured value hai

Example:

URL: student/edit/5

👉 $1 = student/edit/5

🔹 **public/index.php?url=$1**

👉 Ye actual redirect hai

📌 Matlab:

har request ko yahan bhejo:

public/index.php?url=student/edit/5

🔹 **[QSA,L]**

👉 Flags hain

✓ QSA (Query String Append)

👉 Agar URL me already query ho:

?name=ali

👉 to wo bhi preserve rahegi

✓ L (Last)

👉 Ye last rule hai
👉 iske baad aur koi rule apply nahi hoga

🎯 **Full Flow Example**

🔹 **User enters:**

http://localhost/practicproject/student/edit/5

🔸 **Apache check karega:**

✓ file hai? ❌
✓ folder hai? ❌

👉 to rule apply hoga

🔸 **Convert ho jaye ga:**

public/index.php?url=student/edit/5

🔸 **PHP me:**

$_GET['url'] = "student/edit/5";

🔸 **Router:**

['student','edit','5']

🔸 **Final call:**

StudentController->edit(5)

Code Public Index.html -> landing file

```php
<?php

require_once "../app/core/Router.php";

$url = isset($_GET['url']) ? $_GET['url'] : 'students';

$url = explode('/', filter_var(trim($url, '/'), FILTER_SANITIZE_URL));

$controllerName = ucfirst($url[0]) . "Controller";
$method = isset($url[1]) ? $url[1] : "index";
$params = array_slice($url, 2);

require_once "../app/controllers/$controllerName.php";

$controller = new $controllerName();

call_user_func_array([$controller, $method], $params);
```

🔍 **Step-by-Step Explanation**

🔹 **1. <?php**

👉 PHP start ho gaya
👉 Yahan se PHP execute hoti hai

🔹 **2. require_once "../app/core/Router.php";**

👉 require_once = file ko include karo (sirf ek dafa)

📌 Difference:

- require → error pe stop
- require_once → duplicate include nahi hota

👉 Ye Router file load kar raha hai
(although tum abhi manually routing kar rahe ho)

🔹 **3. $url = isset($_GET['url']) ? $_GET['url'] : 'students';**

👉 $_GET['url'] = browser se URL ka part

Example:

http://localhost/practicproject/student/edit/5

👉 .htaccess isko convert karta hai:

public/index.php?url=student/edit/5

Breakdown:

- isset() → check karta hai value exist karti hai ya nahi
- ? : → ternary operator

📌 Meaning:

agar URL hai to use lo, warna default 'students'

🔹 **4. trim($url, '/')**

👉 URL ke start/end se / hata deta hai

/student/edit/5/ → student/edit/5

🔹 **5. filter_var(..., FILTER_SANITIZE_URL)**

👉 URL ko clean karta hai (security)

🔹 **6. explode('/', ...)**

👉 string ko array me convert karta hai

student/edit/5

👉 ban gaya:

$url = ['student', 'edit', '5'];

🔹 **7. $controllerName = ucfirst($url[0]) . "Controller";**

👉 $url[0] = student

👉 ucfirst() → first letter capital

student → Student

👉 final:

StudentController

📌 Matlab:

kaunsa controller load karna hai

🔹 **8. $method = isset($url[1]) ? $url[1] : "index";**

👉 $url[1] = edit

👉 agar method na ho to default:

index

🔹 **9. $params = array_slice($url, 2);**

👉 baqi values parameters ban jati hain

['student','edit','5']

👉 result:

$params = ['5'];

🔹 **10. require_once "../app/controllers/$controllerName.php";**

👉 dynamically file load ho rahi hai

StudentController.php

🔹 **11. $controller = new $controllerName();**

👉 object create ho raha hai

$controller = new StudentController();

🔹 **12. call_user_func_array(...)**

👉 ye dynamic function call karta hai

Breakdown:

[$controller, $method]

👉 matlab:

$controller->edit()

$params

👉 arguments pass ho rahe hain

Final execution:

$controller->edit(5);

Code DB Connection Code

```php
<?php

class Database {
   private $host = "localhost";
   private $db = "test1";
   private $user = "root";
   private $pass = "";

   public function connect() {
      return new PDO("mysql:host=$this->host;dbname=$this->db", $this->user, $this->pass);
   }
}
```

🔍 **Step-by-Step Explanation**

🔹 **1. <?php**

👉 Ye **PHP opening tag** hai
👉 Iske baad jo likha hoga wo PHP code hoga

🔹 **2. class Database**

👉 class ka matlab = **blueprint / template**

👉 Database = class ka naam

📌 Simple words:

Ye ek aisa structure hai jisme hum database ka sara kaam organize karte hain

---

🔹 **3. private $host = "localhost";**

👉 private = ye variable sirf isi class ke andar use ho sakta hai
👉 $host = variable
👉 "localhost" = database server (XAMPP me local machine)

🔹 **4. private $db = "test1";**

👉 Ye tumhara **database name** hai

🔹 **5. private $user = "root";**

👉 MySQL ka username (default = root)

🔹 **6. private $pass = "";**

👉 password (XAMPP me usually empty hota hai)

🔹 **7. public function connect()**

👉 public = ye function bahar se call ho sakta hai
👉 function = method banaya ja raha hai
👉 connect() = function ka naam

📌 Simple:

Ye function database se connection banayega

🔹 **8. return new PDO(...)**

👉 new = naya object create karna
👉 PDO = PHP Data Object (database connect karne ka modern tareeqa)

🔹 **9. "mysql:host=$this->host;dbname=$this->db"**

👉 Ye **connection string** hai

Breakdown:

- mysql: → MySQL database
- host= → server name

- dbname= → database name

🔹 **10. $this->host**

👉 $this = current class object
👉 -> = access operator
👉 $this->host = class ka variable use karna

📌 Matlab:

class ke andar jo $host variable hai usko use karo

🔹 **11. $this->user, $this->pass**

👉 Ye login credentials hain

🎯 **Final Flow (Simple Words)**

1. Tum Database class banate ho
2. connect() function call hota hai
3. PDO database se connect karta hai
4. connection object return hota hai